

The big picture

Guiding question(s)

How do we quantify matter on the atomic scale?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

When you drink a glass of water, you are ingesting molecules of the chemical substance H_2O in its liquid state (**Figure 1**), but how can you know the exact amount of water molecules that end up inside your body? How can we count objects that are so extremely small?

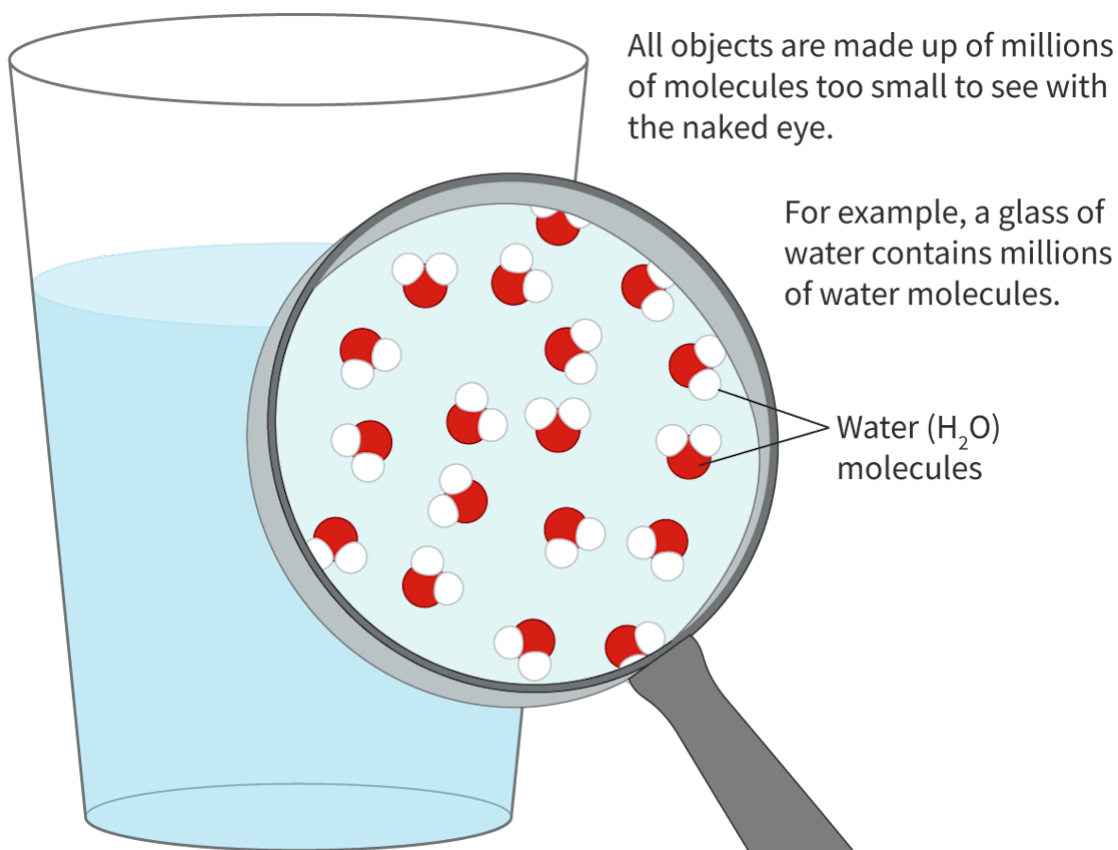


Figure 1. A glass of water contains a huge number of water molecules.

Elementary entities, such as atoms or molecules, are too small to even be seen under a standard electron microscope. Since it is not possible or practical to count these entities, we need to arrange them into large groups and then count the groups using a new unit of measurement called the mole.

Prior learning

Before you study this subtopic make sure that you understand the following:

- The difference between atoms, molecules, ions and subatomic particles (see section S1.1.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>)).
- How to differentiate between relative atomic mass and relative formula mass (see section S1.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>)).
- How to define and give examples of homogeneous mixtures (solutions) (see section S1.1.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>)).

Practical skills

Once you have completed this subtopic, you can practise your laboratory skills with solutions by going to Practical 2a: Preparing a standard solution (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/preparing-a-standard-solution-2a-id-46714/>) and Practical 2b: Determining the concentration of a solution using spectrophotometry/colorimetry (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/determining-the-concentration-of-a-solution-using-spectrophotometrycolorimetry-2b-id-46715/>).